



HAZARD AND RISK

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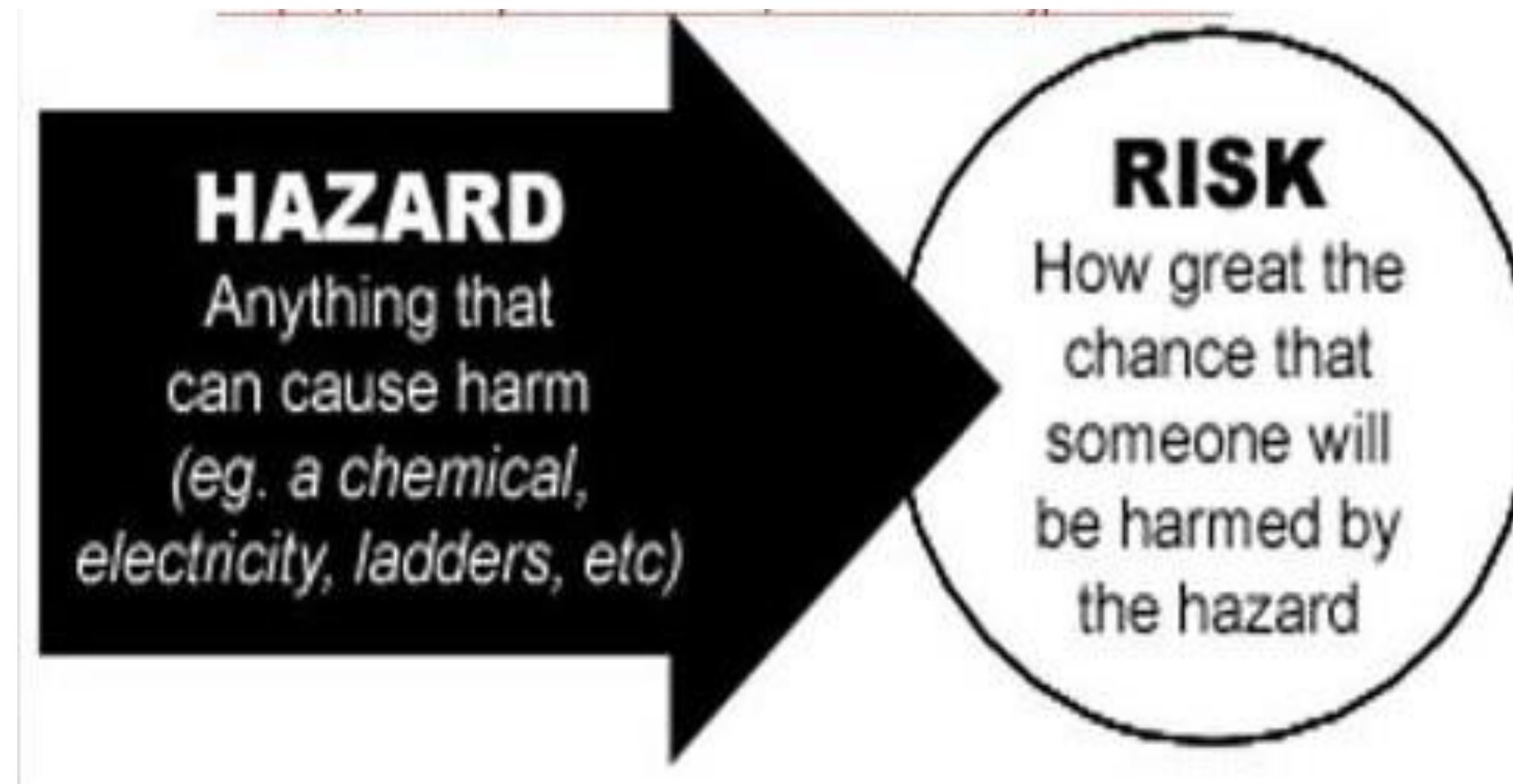
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HAZARD AND RISK

- A hazard is something that can cause harm, e.g. electricity, chemicals, working up a ladder, noise, a bully at work, stress, etc.
- Hazard is a source or a situation with potential to cause harm in terms of human injury or ill-health, damage to property or environment or both.
- Hazards are identified in the performance of various activities, storage and handling of materials and operation and maintenance of equipment's.
- Hazard control is that function which is oriented towards recognizing, evaluating and working towards eliminating hazards and destructive effects found at the workplace.

- A risk is the chance, high or low, that any hazard will actually cause somebody harm.



TYPES OF HAZARDS

Hazards may be classified as under:

- Mechanical Hazards
- Electrical Hazards
- Chemical Hazards.

1. Mechanical Hazards:

- These are responsible for the majority of the accidents in work situations, therefore every workplace and equipment should be properly examined for identifying mechanical hazards and for taking mitigating measures.



Common sources of mechanical hazards are:

- Unguarded or inadequately guarded moving parts or pits etc.
- Machine tools, hand tools, handling materials, lifting and other appliances.
- Improper ventilation, unsafe dress or apparel etc.
- Improper use of tools.

2. Electrical Hazards:

- These may be due to contact of body with wire, cable or rail or from stroke of lightning.
- The immediate effect of this is shock which may be relatively mild or severe so as to cause death (electrocution) depending upon the strength of the current and/or the path it takes passing the earth through the body.
- Another result is burning and the burns may be severe and deep, especially with higher voltage.

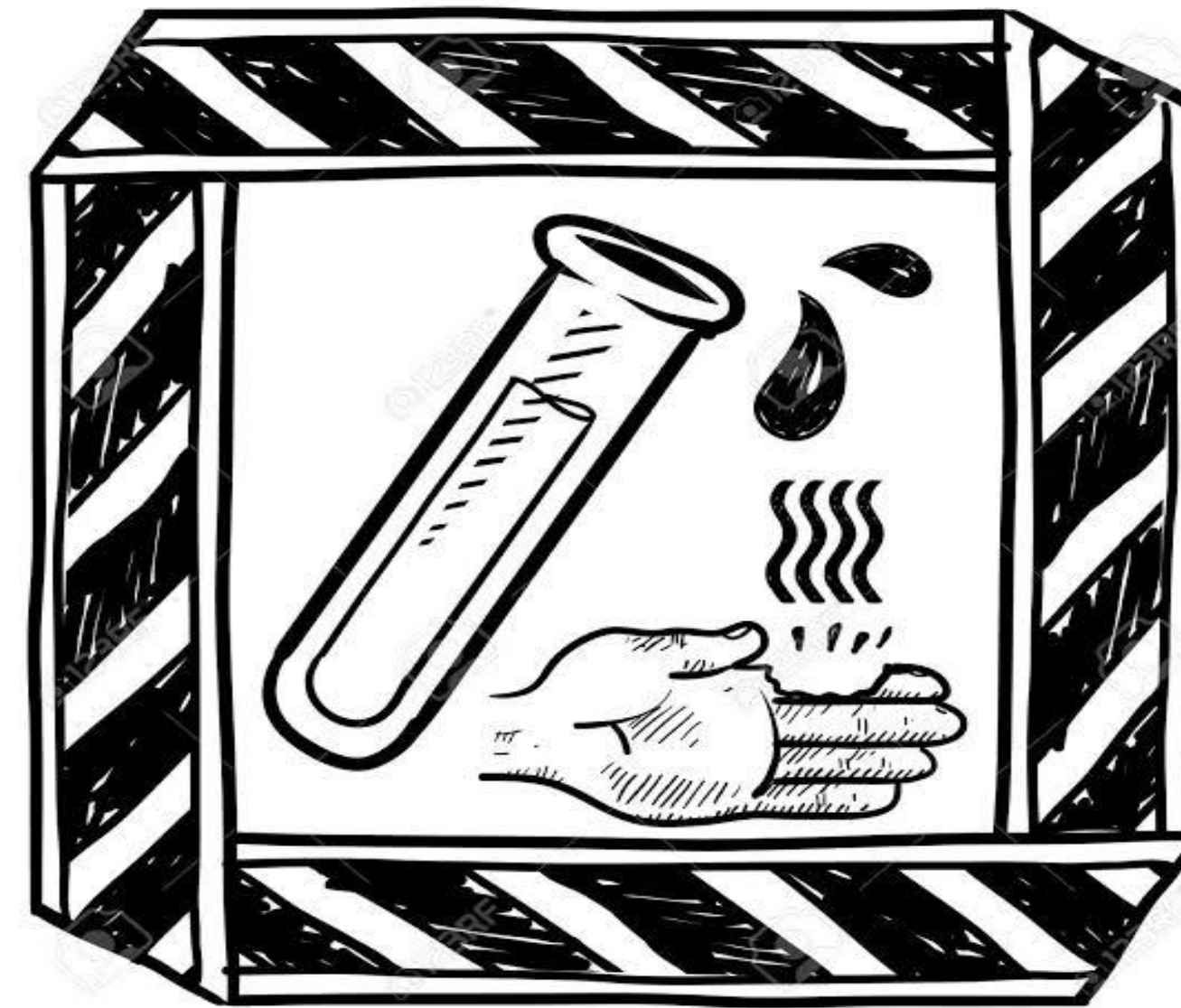


Causes of the electric hazards may be of the following types:

- Electric shocks may be caused by an exposed live conductor or a faulty piece of equipment.
- A mobile crane boom, a man carrying or climbing an aluminium ladder, or vertical metal bars etc. can come in contact with overhead power lines, electric crane rails, open-faced substation switchboards etc.
- Other causes may be unskilled electricians, improper instructions, defective wiring which may cause short circuit, poor installations, misuse or overloading.
- Ageing and attack by foreign materials causes insulation failures which causes electrical fires or cases of electrocution.

3. Chemical hazards

- The usage of chemicals with the resultant hazardous gases, vapours and fumes is one of the most dangerous industries.



The effects of noxious gases are:

- Simple asphyxiants, e.g., nitrogen gas, methane gas, carbon dioxide.
- Chemical asphyxiants, e.g, carbon monoxide, hydrogen-sulphide, hydrocyanic acid.
- Irritant gases, e.g., nitrogen dioxide or peroxide, fluorine, hydrogen fluoride, sulphur dioxide, ammonia.
- Organic metallic gases, e.g., arsenic hydride.
- Inorganic metallic gases.

THANK YOU